National University
of Computer & Emerging Sciences

Program: BS(CS)
Date: 1st May, 2026
Course: Digital Logic Design (DLD)
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Due Date: 4th May, 2026

Assignment #03

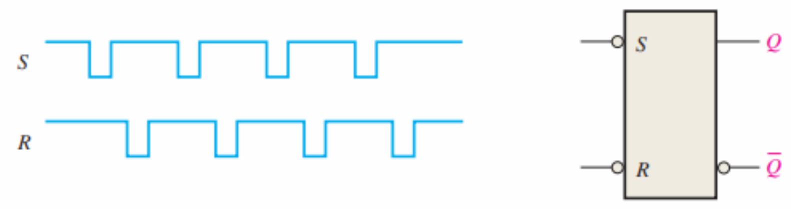
1. There is a mismatch in the given question. Although it mentions an **active HIGH SR latch**, the diagram shows $\bar{S}$ and $\bar{R}$ inputs, which correspond to an **active LOW SR latch** implemented using a NAND gate.

Therefore, you are required to solve the question in **both ways**:

1. First, solve it as an **active LOW SR latch** (based on the given circuit diagram).
2. Then, also draw the waveform (graph) and write the truth table for the **active HIGH SR latch**, which is implemented using a NOR gate, for your conceptual understanding.

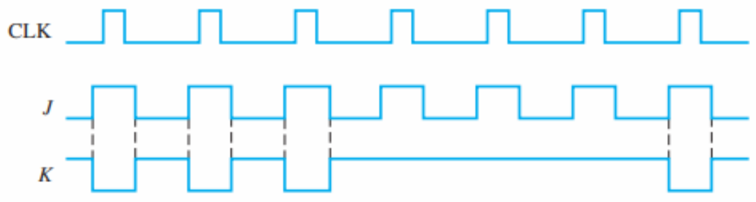
So, you must attempt **both active LOW and active HIGH versions** of the SR latch.

If the waveforms in Figure 7–70 are applied to an active-HIGH S-R latch, draw the resulting Q output waveform in relation to the inputs. Assume that Q starts LOW.



2.

For a negative edge-triggered J-K flip-flop with the inputs in Figure 7–84, develop the Q output waveform relative to the clock. Assume that Q is initially LOW.



3. Design a T Flip-Flop using a JK flip-flop. Explain the conversion process, show the required steps, and draw the final circuit with its truth table.